final-report.md

Final Report — Hiver AI Support Agent (`hulu_support`)

Freeze: brand, taxonomy, golden N=199, splits, retrieval corpus, policy/thresholds.

Headline metric: unsafe auto-handle rate = **6/87** ≈ **0.069** at coverage **33/199** ≈ **0.166** (semantic + policy, golden).

*Meaning: among the 87 gold-escalate cases, 6 were incorrectly auto-handled — **not** 6.9% of all messages.*

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1. Problem framing

Build a trustworthy AI support agent on real Twitter brand conversations: classify intent, draft a reply grounded in historical resolutions, and decide auto-handle vs escalate with a reason. Proof beats demos.

2. What good means for Hulu

Safety-first automation: invent no policy/prices/refunds/account actions; escalate when evidence is weak, conflicting, or account/billing/security-sensitive. Optimize **unsafe auto-handle rate**, not maximum coverage.

3. Scope and non-goals

In scope: `hulu_support` TWCS subsample, 10 intents, retrieve-and-ground drafts, deterministic policy, evaluation.

Not built: live account tools, judge-as-GT labeling, mega-brand scope, Banking77 as primary data, cloud LLM golden eval (no key in freeze).

4. Data

Kaggle TWCS via HF mirror. Brand extract → pairs → conversation split. Train labeled 4000 / valid 800 / golden 199 from held-out test conversations.

5. Leakage prevention

Conversation-level splits; exact duplicate-text purge; retrieval index **train-only**; golden/test IDs blocked; contamination checks at index build.

Demonstrated check (`make check-leakage`):

```
Golden-set leakage check
------------------------
Training overlap:       0
Retrieval overlap:      0
Prompt-example overlap: 0
Status: PASS
```

5b. Evaluation set terminology

Golden N=199 is a **taxonomy-guided human-annotated evaluation set**, not “ground truth.”

Independently validated on a **50-example** second-annotator subset (intent agreement 62%, κ=0.575; escalate 72%, κ=0.435).

5c. Intent vs escalation (final system)

Head	Accuracy	Macro/Esc F1	Notes
Intent	0.693	0.668	TF-IDF+LR
Escalation	0.544	0.640	Policy decision; unsafe FN rate 0.069

By difficulty (intent macro-F1 / escalation F1): easy 0.662 / 0.519; medium 0.600 / 0.698; hard 0.214 / 1.000 (N_hard=9).

Escalation false negatives (dangerous): **6** — see `reports/validation/final_system_metrics.json`.

`other_ambiguous` pred rate 0.261 with **34** false-ambiguous predictions (dump risk).

6. Intent taxonomy

10 human-reviewed intents (playback, live_tv, app/device, login, billing, content, outage, how_to, feedback, other_ambiguous). Discovered on train; reviewed.

7. Baselines

1) Majority intent 2) TF-IDF+LogReg 3) TF-IDF retrieve-and-copy. Semantic agent reuses LR for intent + semantic retrieve + policy.

8. Agent architecture

```
message → TF-IDF+LR intent → semantic retrieve → evidence strength →
deterministic policy → escalate template OR historical copy (LLM optional)
```

9. Retrieval

`tfidf_svd` embeddings; 4000 train cases; diversity (near-dupe + distinct conversations). Golden neighbor intent agree top1 **0.432** / top3 **0.568** / mean sim **0.638** (diagnostic ≠ correctness).

10. Grounded generation

No-LLM: copy top support reply. LLM: OpenAI-compatible JSON drafts from evidence only. Validators flag empty/price/timeline/account-action hallucinations → force escalate.

11. Escalation policy

Final authority. Signals: confidence, similarity, evidence strength/agreement, account/billing/security/ambiguous/unsupported. Thresholds validation-tuned (not golden).

12. Evaluation methodology

Golden N=199 intent+escalation (**human-annotated evaluation set**). Human reply pack N=50 (`annotator_1`). Second-pass intent/escalate N=50 (`annotator_2`) with Cohen's κ. LLM judge `qwen2.5:3b` evaluator-only. Validation suite: `make validate-eval`.

13. Results

System	Acc	Macro-F1	W-F1	Unsafe↓	Coverage	Reply C/G/H (human N=50)
Majority	0.126	0.022	0.028	0.000	0.000	N/A
TF-IDF+LR	0.693	0.668	0.675	0.253	0.317	N/A
TF-IDF retrieval	0.352*	0.362*	0.360*	0.149	0.176	3.14/3.18/3.06
Semantic+policy	0.693	0.668	0.675	**0.069**	0.166	2.92/4.86/2.58
Semantic+LLM	NOT MEASURED					

*neighbor-intent diagnostic. Reply = HUMAN VALIDATED. Semantic actionability/style means 2.58 / 3.02.

14. Top 5 failure modes

1. **Escalation disagreement** — 91/199 (45.7%) = **85 over-escalate (FP)** + **6 under-escalate (FN)**. Ex: `gold_2911030`, `gold_2721090`. Cause: safety-first gates. Mitigation: intent-specific thresholds. *(Flags co-occur with other modes.)*

2. **Intent confusion** — 61/199 (30.7%). Ex: `gold_1372321` playback↔live_tv. Cause: lexical overlap. Mitigation: hierarchical intents.

3. **Rare intents** — 34/199 (17.1%). Ex: `gold_45990` how_to→billing. Cause: low support + price lexicon. Mitigation: more labels / rules.

4. **Ambiguous messages** — 25/199 (12.6%). Ex: thin complaints → other_ambiguous. Cause: insufficient info. Mitigation: clarify prompts (product).

5. **Long multi-issue tweets** — 7/199 (3.5%). Cause: multi-intent. Mitigation: split/detect multi-intent.

15. What is misleading about the headline number?

Headline unsafe auto-handle = 6/87 ≈ 0.069 looks strong. Caveats (quantitative):

- Denominator is **gold-escalate positives (87)**, not all 199 messages.
- Golden N=199 is a **taxonomy-guided human-annotated evaluation set** (not "ground truth"); `annotator_2` subset IAA **62% intent (κ=0.575) / 72% escalate (κ=0.435)** — label ceiling ≠ 100%.
- Low unsafe rate coexists with **coverage only 33/199 ≈ 16.6%** and high escalate rate — safety via refusal; escalation disagreement = **91** (85 over + 6 under).
- Human **helpfulness 2.58 < retrieve-and-copy 3.06** — safer ≠ more helpful.
- Intent **0.693** is the shared LR classifier, not a retrieval win; semantic retrieval changes **evidence + policy**, not the intent model.
- Neighbor intent agree **0.43 ≠ reply correctness**.
- Judge groundedness within±1 **0.12** (v1); must not back the headline.
- Historical ~2017 Twitter ≠ current Hulu policy; temporal drift unmeasured under conversation split.
- Brand suitability heuristic; CAP sensitivity can flip top brand.
- `other_ambiguous` / rule-aid noise in train labels.
- Semantic+LLM reply quality **NOT MEASURED** in freeze.

16. Limitations

- Primary golden labels are taxonomy-guided; full-N independent adjudication is incomplete (IAA on 50/199).
- Rule-aid may introduce confirmation bias (high rule↔gold agreement is partly circular).
- Evaluation set is relatively small (N=199); hard slice has only 9 examples.
- Taxonomy boundaries are subjective (playback vs live_tv).
- Lexical embeddings; uncalibrated confidence still drives many low-conf escalations.
- Historical tweets ≠ current Hulu policy; historical replies are evidence, not escalation truth.
- LLM judge weak on groundedness; Semantic+LLM golden reply quality NOT MEASURED.

17. What I would do with one more week

Intent-specific policy; ST embeddings; LLM draft eval with key; calibrate confidence; expand IAA; near-dupe leakage scan; PII/tooling sketch for production.

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headline_metric.md

Headline metric

Chosen metric

Unsafe auto-handle rate of the frozen semantic retrieval + deterministic policy agent on the golden set.

Value (MEASURED, exact): 6 / 87 = 0.068965… ≈ 0.069

Exact definition

Among golden examples with `gold_escalate = true` (N■ = **87** of 199):

$$
\text{unsafe auto-handle rate} = \frac{\mathrm{FN}}{N^{+}} = \frac{\#\{\text{gold escalate} \land \text{predicted auto-handle}\}}{\#\{\text{gold escalate}\}}
$$

Independent recomputation (frozen agent):

Cell	Count
TP (escalate∧gold escalate)	81
FP (escalate∧gold auto)	85
FN (auto∧gold escalate)	**6**
TN (auto∧gold auto)	27
Gold escalate positives	**87**
Predicted auto-handle	33

- **Unsafe auto-handle** = 6/87 ≈ **0.069**
- **Auto-handle coverage** = 33/199 ≈ **0.166**

- **Escalation recall** = 81/87 ≈ **0.931**
- **Escalation precision** = 81/(81+85) ≈ **0.488**

Critical wording

Do not say “6.9% of all messages are unsafe.”

It is **6.9% of gold-escalate-positive cases** that were incorrectly auto-handled.

At the same operating point, only **16.6% of all golden messages** are auto-handled.

Why chosen

Primary trust risk for a support agent is auto-handling cases that should escalate. Judge groundedness is disqualified (systematic escalation bias). Intent accuracy is shared with TF-IDF+LR.

What it does not measure

Reply helpfulness; over-escalation UX cost; current Hulu policy truth; LLM draft quality.

Source

`reports/final/metric_audit_recompute.json` and `evaluation/results/phase2_evaluation_matrix.json`

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comparison_table.md

Canonical final comparison table

Sources: `evaluation/results/phase2_evaluation_matrix.json`, `reports/phase2/human_reply_summary.json`, human_ratings.csv.

SYSTEM	Intent Acc	Macro F1	W-F1	Intent src	Reply C	Reply G	Reply H	Reply Act	Reply Style	Reply src	Coverage	Esc rate	Unsafe↓	Esc P	Esc R	Decision src
Majority	0.126	0.022	0.028	MEASURED golden199	—	—	—	—	—	NOT MEASURED	0.000	1.000	0.000	0.437	1.000	MEASURED

| TF-IDF+LR | 0.693 | 0.668 | 0.675 | MEASURED | — | — | — | — | — | NOT MEASURED | 0.317 | 0.683 | 0.253 | 0.478 | 0.747 | MEASURED |

| TF-IDF retrieval | 0.352* | 0.362* | 0.360* | MEASURED* neighbor | 3.14 | 3.18 | 3.06 | NOT MEASURED | NOT MEASURED | HUMAN N=50 | 0.176 | 0.824 | 0.149 | 0.451 | 0.851 | MEASURED |

| Semantic retrieval+policy | 0.693 | 0.668 | 0.675 | MEASURED (same LR) | 2.92 | 4.86 | 2.58 | 2.58 | 3.02 | HUMAN N=50 | 0.166 | 0.834 | **0.069** | 0.488 | 0.931 | MEASURED |

| Semantic+LLM | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED | NOT MEASURED |

* Intent for TF-IDF retrieval = neighbor-intent diagnostic (AUTOMATED), not a trained classifier.

Architecture note (Semantic retrieval+policy): intent accuracy matches TF-IDF+LR because the **intent classifier is the same TF-IDF+LogReg**. Semantic retrieval changes **evidence retrieval**; the grounded policy changes **escalation / auto-handle**. Do not treat equal intent metrics as a retrieval win.

LLM judge scores: AUTOMATED evaluator — **not** used as reply headline (groundedness within±1=0.12).

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top5_failure_modes.md

Top 5 failure modes (frozen golden N=199)

Source: `reports/final/metric_audit_recompute.json` + `reports/phase2/failure_analysis.json`.

Overlap: flag categories **co-occur**. Counts are **not mutually exclusive**. An example can contribute to multiple modes.

1. Escalation disagreement — 91 / 199 (45.7%)

Includes **both** directions:

- **Over-escalation (FP):** gold auto, pred escalate = **85**
- **Under-escalation (FN):** gold escalate, pred auto = **6**

Primary safety-relevant subtype for UX/volume is over-escalation (**85**).

Examples (over): `gold_2911030`, `gold_2721090`.

Cause: safety-first policy.

Mitigation: intent-specific auto-handle bands.

2. Intent confusion — 61 / 199 (30.7%)

pred intent ≠ gold intent.

Examples: `gold_1372321` (playback↔live_tv), `gold_2623567`.

Cause: lexical overlap.

Mitigation: hierarchical intents / features.

3. Rare intents — 34 / 199 (17.1%)

gold ∈ {`how_to_feature`,`service_outage`}.

Examples: `gold_45990`, `gold_1138538`.

Cause: low support + lexical bleed into billing.

Mitigation: more rare-class labels.

4. Ambiguous messages — 25 / 199 (12.6%)

gold = `other_ambiguous`.

Cause: insufficient information.

Mitigation: clarify questions (product).

5. Long messages — 7 / 199 (3.5%)

customer_message length > 220.

Cause: multi-issue tweets.

Mitigation: multi-intent detection.

Do **not** label mode #1 as “Over-escalation 91” — that number is total disagreement (85+6).

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case_studies.md

Case studies (10)

Generated from frozen deterministic agent.

normal_successful_auto_or_copy (`gold_2544720`)

- **input:** @115940 when will the new episode of @1760 and the @167381 be on hulu

- **intent:** pred=content_availability (conf=0.780) gold=content_availability
- **evidence_strength:** 0.599; top_sim=0.781
- **reply:** @221742 Sorry for the delay! The new Empire is now up on all supported devices: https://t.co/3hYIXFIT24. Happy streaming!
- **escalation:** False reason=auto_handle_strong_evidence
- **evaluation:** gold_esc=False; pred_esc=False; intent_match=True

escalation_case (`gold_1224206`)

- **input:** Why do I have to pay for Hulu and have to watch commercials? ■■ @115940
- **intent:** pred=billing_subscription (conf=0.689) gold=billing_subscription
- **evidence_strength:** 0.628; top_sim=0.766
- **reply:** I'm escalating this to a Hulu specialist because: billing_refund. A human agent can take a closer look with your account context.
- **escalation:** True reason=billing_refund
- **evaluation:** gold_esc=True; pred_esc=True; intent_match=True

weak_evidence_or_over_escalation (`gold_2911030`)

- **input:** @hulu_support why is the app constantly pausing and buffering lately? It's not connection because it is happening at several different locations and devices for me. From ps4 to Android phone.
- **intent:** pred=playback_error (conf=0.415) gold=playback_error
- **evidence_strength:** 0.481; top_sim=0.480
- **reply:** I'm escalating this to a Hulu specialist because: low_intent_confidence. A human agent can take a closer look with your account context.
- **escalation:** True reason=low_intent_confidence
- **evaluation:** gold_esc=False; pred_esc=True; intent_match=True

ambiguous_case (`gold_199210`)

- **input:** @hulu_support The Purge: Election Year is showing up on my Hulu app but not my tv. It doesn’t say it’s part of the HBO preview weekend. Help
- **intent:** pred=app_device_issue (conf=0.265) gold=other_ambiguous
- **evidence_strength:** 0.465; top_sim=0.543
- **reply:** I'm escalating this to a Hulu specialist because: low_intent_confidence. A human agent can take a closer look with your account context.
- **escalation:** True reason=low_intent_confidence
- **evaluation:** gold_esc=True; pred_esc=True; intent_match=False

billing (`gold_310432`)

- **input:** @hulu_support I signed up for a free trial at the $5.99 rate and switched to $11.99 an hour later and now I'm getting charged $4. Why?
- **intent:** pred=billing_subscription (conf=0.816) gold=billing_subscription
- **evidence_strength:** 0.606; top_sim=0.741
- **reply:** I'm escalating this to a Hulu specialist because: billing_refund. A human agent can take a closer look with your account context.
- **escalation:** True reason=billing_refund
- **evaluation:** gold_esc=True; pred_esc=True; intent_match=True

login_account (`gold_954498`)

- **input:** @hulu_support wait time today to try and figure out why I haven't been able to log in for five days https://t.co/lgt47x96X7
- **intent:** pred=login_account (conf=0.645) gold=login_account
- **evidence_strength:** 0.448; top_sim=0.588
- **reply:** I'm escalating this to a Hulu specialist because: account_specific. A human agent can take a closer look with your account context.
- **escalation:** True reason=account_specific
- **evaluation:** gold_esc=True; pred_esc=True; intent_match=True

playback_live_boundary (`gold_1372321`)

- **input:** @hulu_support ok...I give up. Hulu TV "live" is constantly buffering and is now a solid half inning behind the WS. What do I do?
- **intent:** pred=live_tv_issues (conf=0.342) gold=playback_error
- **evidence_strength:** 0.476; top_sim=0.562
- **reply:** I'm escalating this to a Hulu specialist because: conflicting_evidence. A human agent can take a closer look with your account context.
- **escalation:** True reason=conflicting_evidence
- **evaluation:** gold_esc=False; pred_esc=True; intent_match=False

successful_grounded_reply (`gold_1587983`)

- **input:** @488406 @hulu_support Agreed. I hate the new setup. I hardly ever use it anymore. Almost no reason for me to have it
- **intent:** pred=feedback_complaint (conf=0.787) gold=feedback_complaint
- **evidence_strength:** 0.516; top_sim=0.552
- **reply:** @143922 Sorry for the frustration! On most devices, you can press the back button to get rid of the overlay. We'll share your feedback.
- **escalation:** False reason=auto_handle_strong_evidence

- **evaluation:** gold_esc=True; pred_esc=False; intent_match=True

content_availability (`gold_815865`)

- **input:** @hulu_support Hi! I'm not seeing episode 5, season 6 of @19606 yet. Shouldn't it be available already? Past midnight on a Tuesday.
- **intent:** pred=content_availability (conf=0.833) gold=content_availability
- **evidence_strength:** 0.465; top_sim=0.663
- **reply:** I'm escalating this to a Hulu specialist because: conflicting_evidence. A human agent can take a closer look with your account context.
- **escalation:** True reason=conflicting_evidence
- **evaluation:** gold_esc=False; pred_esc=True; intent_match=True

worst_observed_policy_mismatch (`gold_45990`)

- **input:** @hulu_support I just wanted to know, with the basic package (limited commercials), how many screens can you watch at the same time?
- **intent:** pred=billing_subscription (conf=0.470) gold=how_to_feature
- **evidence_strength:** 0.466; top_sim=0.557
- **reply:** I'm escalating this to a Hulu specialist because: billing_refund. A human agent can take a closer look with your account context.
- **escalation:** True reason=billing_refund
- **evaluation:** gold_esc=False; pred_esc=True; intent_match=False

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EXPERIMENT_FREEZE.md

Experiment freeze

Status: FROZEN for submission

Component	Frozen value
Brand	`hulu_support`
Intent taxonomy	10 intents in `src/intent/taxonomy_hulu.py`
Golden set	`evaluation/golden/golden_set.csv` N=**199**
Splits	conversation-level 70/15/15 in `hulu_support_pairs_split.parquet`

Retrieval corpus	train-only, 4000 cases, `tfidf_svd`
Escalation policy	`GroundedEscalationPolicy`
Thresholds	`data/processed/agent_artifacts/policy_thresholds.json` (validation-tuned)
Eval config	golden N=199; human reply N=50; judge v1 N=50

Do **not** change these to chase metrics. Bug fixes require re-measurement and documentation.

---

performance.json


```
{
  "status": "OBSERVED",
  "n_probe_messages": 15,
  "deterministic_e2e_mean_ms": 27.47,
  "deterministic_e2e_median_ms": 26.68,
  "classifier_mean_ms": 0.86,
  "retrieval_mean_ms": 27.44,
  "index_build_time_note": "Previously observed ~27s for tfidf_svd on 4000 train cases (build_index); not re-run
  "llm_agent_latency": "NOT MEASURED (no cloud API key in freeze run)",
  "llm_judge_runtime": "OBSERVED ~48 minutes for 50 examples (qwen2.5:3b)",
  "targeted_rejudge_runtime": "OBSERVED ~6 minutes for 15 examples (rubric v2)"
}
```


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judge_calibration.md

Judge calibration (Phase 3B)

Status

Item	Value
Judge model	`ollama:qwen2.5:3b`
Pack size	**50/50 complete** (v1 rubric)
Runtime	~**48 minutes** for 50 examples (~1 min/example)
Original scores	`evaluation/reply_human/llm_judge_scores_v1.csv` (also `llm_judge_scores.csv`)
Human reference	`evaluation/reply_human/human_ratings.csv` (`annotator_1`)

Why an LLM judge

Automated ordinal scoring of reply quality at scale, with a fixed rubric and **no gold labels in the prompt**. The judge is an **evaluator**, not ground truth. Human ratings are the reference for calibration.

Rubric

- **v1** (original 50-run): `evaluation/reply_human/judge_rubric_v1.md`
- **v2** (calibrated): `evaluation/reply_human/judge_rubric_v2.md`

v2 adds an explicit rule: **do not penalize groundedness for omitting unsupported details when escalation is the correct behavior.**

Observed agreement (v1 vs human, N=50)

Dimension	Exact	Within ±1	MAE	Weighted κ	Bias (J–H)
correctness	0.36	0.72	—	0.15	–0.66
groundedness	0.06	**0.12**	—	–0.03	–**2.86**
helpfulness	0.32	**0.82**	—	0.21	+0.84
actionability	0.38	**0.88**	—	0.10	+0.14
brand_style	0.06	0.56	—	0.01	+1.38

Full group splits: `reports/phase3/agreement_by_group.json`.

Groundedness root cause (evidence-backed)

Group	n	Human g mean	Judge g mean	Within ±1
Non-escalated	5	3.6	3.2	**1.00**
Escalated	45	5.0	1.87	**0.02**
All	50	4.86	2.0	0.12

Hypothesis: *"judge treats safe escalation as insufficiently grounded."*

- Broad match (escalated ∧ human≥4 ∧ judge≤2): **31** cases
- Share of |diff|>1 disagreements: **~70%**
- **Supported by data** (not assumed a priori)

Top-15 disagreements: `reports/phase3/top15_groundedness_disagreements.json`.

Reliability verdict

- **Strongest:** actionability, helpfulness (within±1)
- **Weakest:** groundedness (v1)
- **Systematic bias:** yes on groundedness for escalated templates (judge under-scores)

Is qwen2.5:3b adequate?

Aspect	Assessment
Helpfulness / actionability screening	Usable with caution
Groundedness (v1 rubric)	**Not adequate** for headline metrics
Runtime	Acceptable for N=50; too slow for frequent full golden runs
Headline metric	**Do not use judge as final headline** — use human ratings + deterministic safety metrics

Targeted re-judge (v2)

Re-run **only** the 15 largest groundedness disagreements with rubric v2 (`qwen2.5:3b`).

Results: `evaluation/reply_human/llm_judge_scores_v2.csv`, `reports/phase3/targeted_rejudge_v2.json`.

Original v1 scores are never overwritten.

Metric on 15-subset	v1	v2
Mean $|diff|$ to human	4.00	**3.07**
Within ±1	0.00	**0.13**
Mean (v2–v1) groundedness	—	**+0.93**

Interpretation: Rubric v2 moves scores in the right direction on average but does **not** fully repair groundedness agreement on this 3B model. Headline recommendation unchanged: do not use judge groundedness as product GT.

What human validation would improve

1. Blind re-rate groundedness on escalated templates with the v2 definition.
2. Separate “grounded abstention” from “grounded answer” in the human form.

3. Larger N and a second annotator for IAA on reply dimensions.

Non-negotiable

> **LLM judge = evaluator. Human ratings = reference. Judge must not label the test/golden set as ground truth.**

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decision-log-short.md

Decision log (final) — 12 decisions

1. **Decision:** Explicit TWCS acquisition (no silent full download).

Why: Auditability / offline safety.

Alternative: Auto-download on import.

Trade-off: Extra setup step.

Consequence: Reproducible installs without surprise 500MB pulls.

2. **Decision:** Brand = `hulu_support` by suitability, not max volume.

Why: Cleaner 6–12 intent taxonomy.

Alternative: AmazonHelp.

Trade-off: Less data than mega-brands.

Consequence: Focused evaluation narrative.

3. **Decision:** Conservative conversation IDs for huge multi-author components.

Why: Undirected graphs merged unrelated customers.

Alternative: Always trust components.

Trade-off: Possible over-fragmentation.

Consequence: Safer splits.

4. **Decision:** Conversation-level split + exact duplicate-text purge.

Why: Prevent thread/text leakage.

Alternative: Tweet-level random split.

Trade-off: Slightly smaller eval pools.

Consequence: Honest retrieval/intent metrics.

5. **Decision:** 10 human-reviewed intents from train-only discovery.

Why: Fits Hulu themes without Banking77 sprawl.

Alternative: Raw cluster IDs or 77 banking intents.

Trade-off: Boundary fuzz (playback vs live).

Consequence: Interpretable taxonomy.

6. **Decision:** Keep TF-IDF+LR as primary intent classifier.

Why: Measured baseline; embeddings ≠ automatic upgrade.

Alternative: Embed+linear or LLM classify.

Trade-off: Lexical limits.

Consequence: Semantic agent intent = 0.693/0.668 matches LR.

7. **Decision:** Default embeddings = `tfidf_svd`.

Why: CI/demo without model downloads.

Alternative: Force sentence-transformers.

Trade-off: Weaker paraphrase retrieval.

Consequence: <15 min headline path.

8. **Decision:** Deterministic escalation policy is final authority.

Why: LLM must not override safety.

Alternative: Model-decided escalate.

Trade-off: Over-escalation.

Consequence: Unsafe auto-handle 0.069 at coverage 0.166.

9. **Decision:** Tune thresholds on validation only.

Why: Golden integrity.

Alternative: Tune on golden.

Trade-off: Noisy validation labels.

Consequence: Honest held-out safety metric.

10. **Decision:** Prefer unsafe auto-handle minimization over coverage.

Why: Trust > automation rate.

Alternative: Maximize auto-handle.

Trade-off: Helpfulness lower on human pack.

Consequence: Headline = unsafe 0.069.

11. **Decision:** LLM judge = evaluator only; not GT.

Why: Groundedness within±1 = 0.12 under v1; esc bias.

Alternative: Use judge scores as labels.

Trade-off: Need human pack for reply quality.

Consequence: Reply headlines from annotator_1.

12. **Decision:** Freeze brand/taxonomy/splits/policy for submission.

Why: Stop metric chasing.

Alternative: Keep iterating thresholds on golden.

Trade-off: Known over-escalation remains.

Consequence: Stable submission package.

SUBMISSION_CHECKLIST.md

Submission checklist

- [x] runnable repository
- [x] README complete (reviewer quickstart)
- [x] golden set 150–250 (N=199)
- [x] evaluation harness
- [x] baseline comparison (≥2)
- [x] reply evaluation (human N=50)
- [x] escalation evaluation (golden)
- [x] LLM judge rubric (v1 + v2)
- [x] judge analysis (Phase 3B)
- [x] five failure modes
- [x] misleading headline section
- [x] decision log (12)
- [x] six-page report (`reports/final-report.md`)
- [x] interview PDF (`make docs`)
- [x] no secrets in repo (`.env` gitignored)
- [x] no accidental huge files guidance (raw TWCS gitignored)
- [x] tests pass
- [x] deterministic demo works

Reviewer commands

```
python -m src.cli.build_index --embedding tfidf_svd
python -m src.cli.eval_phase2
python -m src.cli.run_agent --mode deterministic --message "Hulu keeps buffering when I try to watch live TV"
make test
make docs
```

Submit

Form: https://intelligent-bar-256.notion.site/39492cbf0da2800682cfc78a600a745f?pvs=105

Include public/private repo link + `reports/final-report.md` / PDF.

---

project-engineering-guide.md

Hiver AI Support Agent — Project Engineering Guide

Phase covered: Phase 0 → 2 (grounded `hulu_support` agent)

PDF regeneration: `make docs` or `python -m src.cli.generate_docs`

Status tags: IMPLEMENTED | PLANNED | OBSERVED | ASSUMED | NOT YET MEASURED

---

1. Project overview

Build an AI customer-support agent on Kaggle TWCS for locked brand `hulu_support`.

Given a customer message the system must:

1. Classify intent (reviewed 10-intent taxonomy)
2. Retrieve historically similar support cases (train-only)
3. Draft a reply grounded in evidence (or escalate)
4. Decide auto-handle vs escalate with an explicit reason
5. Expose evidence + confidence

Critical rule: The LLM is **not** the source of truth. Historical support evidence is. Weak/conflicting evidence → **ESCALATE**.

---

2. Architecture (Phase 2)

```
Customer message
       ■
       ▼
■■■■■■■■■■■■■■■■■■■■■
■ TF-IDF + LogReg  ■  predicted_intent, intent_confidence (uncalibrated)
■■■■■■■■■■■■■■■■■■■■■
         ▼
■■■■■■■■■■■■■■■■■■■■■
■ Semantic retrieve■  train-only corpus; diversity; top_k hits
■ (tfidf_svd emb.) ■
■■■■■■■■■■■■■■■■■■■■■
         ▼
■■■■■■■■■■■■■■■■■■■■■
■ Evidence strength■  top_sim, gap, intent agree, n_above, reply consistency
■■■■■■■■■■■■■■■■■■■■■
         ▼
■■■■■■■■■■■■■■■■■■■■■
■ Escalation policy■  DETERMINISTIC FINAL AUTHORITY (LLM cannot override)
■■■■■■■■■■■■■■■■■■■■■
         ■
    escalate?■■yes■■■ escalation template + reason + evidence
         ■
         no
         ▼
■■■■■■■■■■■■■■■■■■■■■
■ Reply generator  ■  deterministic copy  OR  grounded LLM JSON draft
■■■■■■■■■■■■■■■■■■■■■
         ▼
■■■■■■■■■■■■■■■■■■■■■
■ Generation valid.■  empty / price / timeline / account-action flags → force escalate
■■■■■■■■■■■■■■■■■■■■■
         ▼
   AgentResponse JSON
```

Components: `src/agent/grounded_agent.py`, `src/retrieval/*`, `src/policy/grounded_policy.py`, `src/generation/providers.py`

---

3. Data flow & reproducibility path

```
# 1 install
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt

# 2 configure
cp .env.example .env   # set DATA_PATH if needed; brand already hulu_support

# 3 build retrieval index (train only; ~30s on labeled train)
python -m src.cli.build_index --embedding tfidf_svd

# 4 evaluate (golden N=199)
python -m src.cli.eval_phase2

# 5 deterministic demo (no API key)
```

```
python -m src.cli.run_agent --mode deterministic \
  --message "Hulu keeps buffering when I try to watch live TV"

# 6 optional LLM
# export OPENAI_API_KEY=... ; python -m src.cli.run_agent --mode llm --message "..."
# python -m src.cli.eval_phase2 --llm

# 7 docs
make test && make docs
```

---

4. Retrieval

Item	Value
Brand	`hulu_support` (unchanged)
Corpus	TRAIN labeled only (4000 cases)
Embedding default	`tfidf_svd` (TF-IDF → TruncatedSVD, L2)
Optional	`sentence_transformers` / `all-MiniLM-L6-v2`
Diversity	near-dupe dedupe + prefer distinct conversations
Contamination	golden/test IDs blocked; check at build

Trade-off: diversity may skip a slightly nearer duplicate from the same conversation in favor of independent evidence.

OBSERVED retrieval quality (golden): index size **4000**; top-1 neighbor intent agree **0.432**; top-3 **0.568**; mean similarity **0.638**. These are **diagnostics**, not reply correctness.

Architecture note: `semantic_retrieval_no_llm` reuses the **same TF-IDF+LR intent classifier** as `tfidf_logreg`. Semantic retrieval changes evidence; policy changes escalation. Equal intent metrics are expected.

---

5. Escalation policy

Inputs: intent confidence, top similarity, evidence strength, evidence agreement, account/billing/security/unsupported/ambiguity signals.

Reasons: `low_intent_confidence`, `weak_evidence`, `conflicting_evidence`, `account_specific`, `billing_refund`, `security_privacy`, `ambiguous_request`, `unsupported_operation`, `potentially_harmful_automation`.

Thresholds (validation-tuned, NOT golden):

- intent_confidence_min: **0.55**
- top_similarity_min: **0.15**
- evidence_strength_min: **0.30**
- auto_handle_similarity_min: **0.30**
- auto_handle_evidence_min: **0.40**

Artifact: `data/processed/agent_artifacts/policy_thresholds.json`

---

6. Generation

- **Deterministic / no-LLM:** copy top historical support response (or escalation template).
- **LLM:** OpenAI-compatible provider; structured JSON; grounded prompt forbids inventing policy/prices/refunds.
- **Validation:** empty, hallucinated price/timeline/account action → force escalate.

---

7. Evaluation matrix (OBSERVED)

Golden N=199. Reply C/G/H from human ratings pack (n=50, `annotator_1`) where noted.

System	Intent acc	Macro F1	Unsafe auto-handle	Coverage	Reply C/G/H
majority	0.126	0.022	0.000	0.000	NOT YET MEASURED
tfidf_logreg	0.693	0.668	0.253	0.317	NOT YET MEASURED
tfidf_retrieval	0.352	0.362	0.149	0.176	**3.14 / 3.18 / 3.06 (human)**
semantic_retrieval_no_llm	0.693	0.668	**0.069** (6/87)	0.166 (33/199)	**2.92 / 4.86 / 2.58 (human)**
semantic_llm	NOT MEASURED				NOT MEASURED

Full JSON: `evaluation/results/phase2_evaluation_matrix.json`

---

8. HUMAN EVALUATION COMPLETED — READ THIS

Annotators

Field	Value
Reply rater	`annotator_1`
Second intent/escalate rater	`annotator_2`
Pack size	50 examples each

Work item A — Reply human pack (50 examples)

File: `evaluation/reply_human/human_ratings.csv`

Fields filled:

- `human_correctness` / `human_groundedness` / `human_helpfulness` → scores of **semantic Phase-2 agent** reply (1–5)
- `human_escalation` → whether a specialist should handle (`true`/`false`)
- `human_comments` → short rationale
- `baseline_correctness` / `baseline_groundedness` / `baseline_helpfulness` → same scales for TF-IDF retrieve-and-copy
- `annotator_id` = `annotator_1`

Means (OBSERVED):

System	Correctness	Groundedness	Helpfulness
TF-IDF retrieve-and-copy	3.14	3.18	3.06
Semantic deterministic agent	2.92	**4.86**	2.58

Escalation:

- Semantic used escalate-template on **90%** of the 50-pack
- Human would escalate **54%**
- Decision agreement semantic vs human: **60%**

Interpretation: Semantic is very grounded (templates invent no policy) but over-escalates easy content/playback FAQs, hurting correctness/helpfulness vs a good historical copy.

Work item B — Second annotator pack (50 examples)

File: `evaluation/golden/second_annotator.csv`

Filled by `annotator_2`: `gold_intent_2`, `gold_escalate_2`, `annotator2_notes` for all 50 rows.

Agreement vs original gold:

- Intent exact agreement: **62%**
- Escalate exact agreement: **72%**

Work item C — Write-ups

- `reports/phase2/HUMAN_ANNOTATION.md`
- `reports/phase2/human_reply_summary.json`
- `reports/phase2/second_annotator_agreement.json`

LLM-as-judge ↔ human agreement: **NOT YET MEASURED** (requires API judge run on the same pack).

---

9. Failure modes (OBSERVED)

Top flags on golden semantic agent (`reports/phase2/failure_analysis.json`):

1. Escalation disagreement (often over-escalate)
2. Intent confusion
3. Rare intents (`service_outage`, `how_to_feature`)
4. Ambiguous messages

---

10. Tests & CLI

- **Tests:** 51 passed (includes Phase-2 contamination, ordering, policy, no-LLM, mocked LLM)
- **Demo:** `python -m src.cli.run_agent --mode deterministic --message "..."`

Example observed demo:

- Intent `live_tv_issues` conf≈0.91
- Evidence strength≈0.53
- Auto-handle with copied historical buffering reply

---

11. Known limitations

- `tfidf_svd` is lexical, not deep semantic
- Classifier confidence uncalibrated
- Generation validators incomplete
- Historical tweets ≠ current Hulu policy
- LLM path not measured without API key
- Judge↔human agreement not yet measured

---

12. Interview questions (short)

1. Why isn’t the LLM the source of truth?
2. Why tune thresholds on validation not golden?
3. Why is semantic groundedness high but helpfulness low in the human ratings?
4. How do you prove retrieval isn’t contaminated by golden?

See `reports/final-report.md` and `reports/decision-log-short.md` for interview-facing summaries.

HOW WE VALIDATED THE EVALUATOR (Phase 3B)

Methodology

- 50 human-rated reply examples (`annotator_1`)
- LLM judge: `ollama:qwen2.5:3b`, fixed rubric, **no gold labels**
- Runtime ≈ 48 minutes for 50 examples

Agreement (v1)

- Helpfulness within±1: **0.82**
- Actionability within±1: **0.88**
- Groundedness within±1: **0.12**

Groundedness problem

Concentrated in **escalated** replies (within±1 ≈0.02) vs non-escalated (1.00). Judge mean groundedness on escalations ≈1.87 vs human 5.0.

Escalation analysis

Hypothesis that the judge treats safe escalation as ungrounded covers ~70% of large disagreements — **supported**.

Limitations

Judge ≠ ground truth. Do not label golden/test with the judge. Prefer human ratings + unsafe auto-handle for headlines.

Recommended next step

Adopt rubric v2; confirm with targeted human re-rate of escalated groundedness; optional larger N.

See `reports/phase3/judge_calibration.md` and `reports/final-report.md`.

HOW TO EXPLAIN THIS PROJECT IN AN INTERVIEW

60 seconds

Hulu Twitter support agent: classify → retrieve train-only history → policy escalate/auto-handle. Headline unsafe auto-handle **6.9%** at 16.6% coverage. Proof via golden 199 + human ratings; LLM judge is evaluator only.

3 minutes

Cover brand choice, leakage-safe splits, TF-IDF+LR vs retrieve-and-copy vs semantic agent, safety-first policy, over-escalation trade-off, judge groundedness failure on escalate templates.

Demo command

```
python -m src.cli.run_agent --mode deterministic --message "Hulu keeps buffering when I try to watch live TV"
```

If asked "what's misleading?"

Unsafe 6.9% looks great partly because we refuse most cases; helpfulness is lower than retrieve-and-copy; labels are single-annotator; historical tweets ≠ current policy.

Repo map

`src/agent` pipeline · `src/retrieval` · `src/policy` · `evaluation/golden` · `reports/final-report.md`

---

engineering-notes.md

Engineering Notes (living document)

> Audience: a technically competent interviewer/reviewer who did not write this code.

> Update after every meaningful implementation step.

> Status tags used below: **IMPLEMENTED | PLANNED | OBSERVED | ASSUMED | NOT YET IMPLEMENTED**

Phase 0 — Audit, data foundation, architecture

1. What was changed

Scaffolded a runnable Phase 0 repository from an empty GitHub repo (`README.md` only) into a structured ML/SDE project with:

- configuration (`configs/default.yaml`, `src/config.py`, `.env.example`)
- dataset acquisition interface (no auto full-download)
- schema validation + loader
- reproducible audit + brand ranking CLIs
- conversation reconstruction utilities
- leakage-aware conversation/temporal splits
- intent discovery proposal pipeline (TF-IDF + KMeans)
- baselines (majority; TF-IDF+LogReg; retrieve-and-copy replies)
- escalation policy with explicit reasons
- evaluation harness + metrics interfaces
- LLM-judge interface (no backend yet)
- golden-set sampling design (no labels yet)
- tests, Makefile, PDF doc generator

2. Why it was changed

The assignment prioritizes evaluation and proof over a flashy demo. Jumping to an LLM agent before understanding schema, brand suitability, conversation structure, and leakage would create non-reproducible, non-interviewable work.

3. How it works (end-to-end Phase 0)

1. User places `sample.csv` / `twcs.csv` under `data/raw/` or sets `DATA_PATH`.
2. `python -m src.cli.main audit` writes `reports/phase0/data_audit.json`.
3. `... brands` ranks outbound brand handles and recommends one from evidence.
4. `... conversations` rebuilds threads and customer→support pairs.
5. `... intents` clusters inbound texts into a **proposal** taxonomy.
6. `... golden-design` writes the labeling methodology artifact.
7. `make test` validates core invariants on a synthetic fixture.
8. `make docs` regenerates the interview PDF from Markdown sources.

4. Important implementation details

Dataset acquisition (**IMPLEMENTED** interface; full corpus **NOT OBSERVED** locally)

- Environment had **no** local Kaggle dump and **no** `~/.kaggle/kaggle.json`.
- Per project rules, we did **not** auto-download the multi-million-row file.
- `src/data/acquire.py` documents manual download and optional explicit CLI download.
- Unit tests use `tests/fixtures/mini_twcs.csv` (**synthetic**, schema-compatible).
- Any JSON under `reports/phase0/` generated against that fixture is tagged `SYNTHETIC_TEST_FIXTURE` and must not be presented as Kaggle findings.

Schema (**ASSUMED** from Kaggle docs, **validated** at load)

Expected columns:

`tweet_id, author_id, inbound, created_at, text, response_tweet_id, in_response_to_tweet_id`

IDs are coerced to strings to avoid float artifacts. `response_tweet_id` may be comma-separated.

Brand identification (**ASSUMPTION**)

Outbound (`inbound == False`) authors whose `author_id` is not purely numeric are treated as brand handles. This follows Kaggle documentation that customers are anonymized numeric IDs.

Conversation reconstruction (**IMPLEMENTED**)

- Build undirected connected components from `in_response_to_tweet_id` edges and in-file `response_tweet_id` edges.

- Missing parents → partial trees with `has_missing_parent=True`.
- Customer→support pairs: outbound tweet whose parent is inbound.
- `first_support_only=True` keeps earliest support reply per customer tweet.

Leakage strategy (**IMPLEMENTED** utilities)

- Default split: **conversation-level** random split (seeded).
- Alternative: **temporal** split by earliest timestamp.
- Contamination checkers ensure golden/test IDs never enter train or retrieval index.

Why not tweet-level random split? The same conversation (and near-duplicate canned replies) would leak across train/test and inflate retrieval + intent metrics.

Intent discovery (**IMPLEMENTED** proposal only)

Phase 0 uses TF-IDF + KMeans (light deps, deterministic seed). Output is explicitly `PROPOSAL_NOT_GROUND_TRUTH`. Embedding models are configured but **NOT YET IMPLEMENTED** for clustering.

Baselines (**IMPLEMENTED** interfaces/classes; not evaluated on real golden labels yet)

- Baseline 0: majority intent
- Baseline 1: TF-IDF + Logistic Regression
- Reply baseline: TF-IDF cosine retrieve → copy historical support response

Escalation (**IMPLEMENTED** conservative rules)

Escalate on low confidence/evidence, account/billing/security language, ambiguity, conflicting evidence, missing info. Auto-handle only with strong positive evidence and no risk signals. Always returns a reason.

LLM judge (**PLANNED** / interface **IMPLEMENTED**)

`src/evaluation/judge.py` defines inputs/outputs and prompt builder. Gold labels are excluded from judge input by design. Agreement metrics are specified but **not computed** (no human annotations yet).

5. Commands used

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
make test
DATA_PATH=tests/fixtures/mini_twcs.csv make phase0   # smoke only
make docs
python -m src.cli.main acquire --instructions
```

6. Files created/modified (Phase 0)

- `src/**` — library + CLI
- `configs/default.yaml`, `.env.example`, `requirements.txt`, `Makefile`, `pytest.ini`
- `tests/**` including synthetic fixture
- `reports/engineering-notes.md`, `reports/project-engineering-guide.md`
- `README.md` updated
- `data/`, `evaluation/`, `notebooks/` structure

7. Alternatives considered

Decision	Alternative	Why not chosen now
TF-IDF clustering	sentence-transformers embeddings	Heavier deps; Phase 0 prioritizes foundation + reproducibility
Conversation split	Tweet-level random split	Causes conversation/retrieval leakage
Auto-download full TWCS	Explicit acquisition	Repo rule + avoid surprise multi-hundred-MB download
Hardcode SpotifyCares	Evidence-based ranking	Assignment says Spotify is only a hypothesis
Build full LLM agent	Baselines + interfaces first	Evaluation quality before system complexity

8. Why the selected approach was chosen

It creates an interviewable, testable spine: data → conversations → leakage-safe splits → intent proposal → baselines → metrics/policy. Every claim can be traced to code or marked as not yet observed.

9. Risks / limitations

- Mention-based inbound brand detection undercounts when `@Brand` was anonymized to numeric IDs.
- TF-IDF clustering can create incoherent “intents”; human review is mandatory.
- Conservative escalation will reduce auto-handle coverage (intentional).
- Timestamp parsing of Twitter date strings can be slow/format-ambiguous at full scale.

10. What to explain in an interview

- How conversations are reconstructed and which edge cases exist
- Why conversation-level splits matter for retrieval systems
- How golden-set contamination would silently invalidate the take-home
- Why baselines exist before LLMs
- Why escalation reasons are first-class outputs
- What is implemented vs merely designed

Phase 1 — Real data foundation (completed)

See also `reports/engineering-notes-phase1.md` for the detailed Phase 1 log.

Summary of what changed

1. **Acquired real TWCS** (`data/raw/twcs.csv`, 2,811,774 rows) via explicit HuggingFace mirror workflow.
2. **Selected** `hulu_support` from evidence (`evaluation/brand_selection.csv`), not max volume / not hardcoded Spotify.
3. **Validated reconstruction**; introduced conservative conversation IDs after observing 232-turn multi-customer mergers.
4. **Built brand pairs dataset** with inference-safe context (no future support text).
5. **Conversation-level splits** + duplicate-text cleanup; temporal split compared.
6. **Human-reviewed 10-intent taxonomy** for Hulu; train-only discovery used as hints only.
7. **Golden set N=199** labeled under engineer protocol + review; isolated from train/retrieval.
8. **Baselines evaluated on golden** with real metrics saved under `evaluation/results/`.
9. **Decision log** + docs/PDF updated.

Observed baseline headline numbers (golden, N=199)

- Majority accuracy ≈ **0.126**, macro-F1 ≈ **0.022**
- TF-IDF+LogReg accuracy ≈ **0.693**, macro-F1 ≈ **0.668**, weighted-F1 ≈ **0.675**
- Retrieve-and-copy mean top-1 cosine ≈ **0.310**; neighbor same-intent rate ≈ **0.352** (retrieval diagnostic only — not reply correctness)

Phase 2 next step

Build the grounded reply + escalation agent (still evaluation-first): retrieval improvements, optional LLM drafting with evidence constraints, human reply ratings, calibrated judge — without contaminating golden/retrieval.

Phase 1.5 — Evaluation Integrity Review

Status tags: **IMPLEMENTED | OBSERVED | KNOWN LIMITATION | NOT YET VALIDATED**

What changed

1. **Golden labeling honesty:** taxonomy-guided human-annotated evaluation set (N=199); not “ground truth.” Second-annotator subset IAA completed (intent 62% / κ=0.575; escalate 72% / κ=0.435). See `reports/validation/`.

2. **Escalation rubric (IMPLEMENTED):** policy judgments with explicit reason codes; not historical Hulu escalation.

3. `other_ambiguous` **audit (OBSERVED):** documented as conceptual `ambiguous_or_other` = insufficient information for one supported intent; not renamed in data (labels preserved). Frequencies + confusion reported.

4. **Brand score audit (OBSERVED):** exact formula, weights, cap normalization, mega-brand penalty, sensitivity. Hulu robust under original CAP=5000; **not** robust if CAP raised (Spotify/Uber/Amazon can lead).

5. **Conversation size audit (OBSERVED):** percentiles + largest components; multi-customer merges confirmed; conservative split IDs remain the modeling choice.

6. **Leakage re-check (OBSERVED):** all integrity checks **PASS**.

7. **Baselines re-run (OBSERVED):** clean invocation + execution manifest; metrics reproduced (not hardcoded).

8. **Second-annotator package (IMPLEMENTED / OBSERVED):** 50 labeled examples; intent agreement 62% (κ=0.575), escalate 72% (κ=0.435). See `reports/validation/second_annotator_iaa.json`.

Artifacts

- \`reports/phase1_5/*\`
- \`evaluation/golden/ESCALATION_RUBRIC.md\`
- \`evaluation/golden/second_annotator.csv\`
- \`evaluation/results/intent_baseline_results.json\` (reproduced)
- \`reports/phase1_5/baseline_execution_manifest.json\`

Phase 2 — Grounded support agent (\`hulu_support\`)

Status tags: **IMPLEMENTED | OBSERVED | NOT YET MEASURED**

1. What was changed

Built the end-to-end grounded agent behind existing abstractions:

- Train-only retrieval corpus + contamination checks (\`src/retrieval/corpus.py\`)
- Embedding backend abstraction: \`tfidf_svd\` (default/CI) + optional \`sentence_transformers\` (\`src/retrieval/embeddings.py\`)
- Semantic retriever with diversity + explicit evidence-strength formula (\`src/retrieval/semantic.py\`)
- Deterministic escalation policy as final authority (\`src/policy/grounded_policy.py\`); thresholds tuned on **validation**, not golden

- Provider-abstracted reply generation + post-generation validation (`src/generation/providers.py`)
- `GroundedSupportAgent` (`src/agent/grounded_agent.py`) modes: `deterministic` | `llm`
- CLIs: `build_index`, `run_agent`, `eval_phase2`
- Evaluation matrix, retrieval quality, failure analysis, ~50 human reply pack, LLM-judge implementation
- Tests: `tests/test_phase2_agent.py` (mocked LLM)

2. Why

The LLM is not the source of truth. Historical support evidence grounds drafts; weak/conflicting evidence → escalate. Safety and groundedness beat automation coverage.

3. How (data flow)

```
customer message
  → TF-IDF+LogReg intent (+ confidence, uncalibrated)
  → semantic retrieve(top_k) from TRAIN corpus only
  → evidence_strength(top_sim, gap, intent agreement, n_above, reply consistency)
  → GroundedEscalationPolicy (deterministic; LLM cannot override)
  → if escalate: template + reason
    else: DeterministicCopyGenerator OR OpenAI-compatible grounded JSON draft
         → validate_generation (prices/timelines/account actions/empty)
         → force escalate on validation flags
  → AgentResponse JSON (intent, evidence, draft, escalate, signals)
```

4. Retrieval

- **Embedding (OBSERVED default):** `tfidf_svd` (TF-IDF → TruncatedSVD, L2-normalized). Optional ST model documented but not required for CI/demo.
- **Index size (OBSERVED):** 4000 train cases after contamination filter; diversity dedupes near-identical messages and prefers distinct conversations at query time.
- **Never indexed:** golden/test conversations and tweet IDs.
- **Trade-off:** nearest-neighbor relevance vs diversity — diversity may skip a slightly higher-scoring near-duplicate from the same thread in favor of independent cases.

5. Escalation thresholds (OBSERVED, validation-tuned)

Saved in `data/processed/agent_artifacts/policy_thresholds.json`:

- intent_confidence_min: 0.55
- top_similarity_min: 0.15
- evidence_strength_min: 0.30
- auto_handle_similarity_min: 0.30
- auto_handle_evidence_min: 0.40

Rationale: minimize unsafe auto-handle on validation labels; secondary coverage. Golden unused for tuning.

6. Evaluation (OBSERVED on golden N=199)

See `reports/phase2/evaluation_matrix.md`. Headline:

System	Intent acc	Macro F1	Unsafe auto-handle	Coverage
majority	0.126	0.022	0.000	0.000
tfidf_logreg	0.693	0.668	0.253	0.317
semantic_retrieval_no_llm	0.693	0.668	0.069	0.166
semantic_llm	NOT YET MEASURED			

Reply quality (correctness/groundedness/helpfulness/…): **NOT YET MEASURED** (needs human ratings).

Retrieval diagnostics (not reply correctness): index **4000**; top-1 neighbor intent agree **0.432**; top-3 **0.568**; mean similarity **0.638**.

7. Human eval / judge

- Pack: `evaluation/reply_human/human_ratings.csv` (**filled** by `annotator_1` — see `reports/phase2/HUMAN_ANNOTATION.md`)
- Second annotator: `evaluation/golden/second_annotator.csv` (**filled** by `annotator_2`)
- Judge: `OpenAICompatibleJudge` **IMPLEMENTED**; judge↔human agreement **NOT YET MEASURED** (no API judge run on pack)

8. Limitations

- TF-IDF SVD is lexical; synonymy/paraphrase weaker than true semantic embeddings.
- Classifier confidence is not calibrated.
- Rule-based generation validation is incomplete by design.
- Historical tweets ≠ current Hulu policy.
- Escalation disagreements with golden often = over-escalate (safety-first), not under-escalate.

9. Commands

```
python -m src.cli.build_index --embedding tfidf_svd
python -m src.cli.eval_phase2
python -m src.cli.run_agent --mode deterministic --message "..."
# optional: export OPENAI_API_KEY=... ; --mode llm / eval_phase2 --llm
make test && make docs
```

Phase 3B — LLM judge calibration

OBSERVED: 50/50 scores with `ollama:qwen2.5:3b` (~48 min). Helpfulness/actionability within±1 high; groundedness 0.12.

Root cause (supported): judge under-scores safe escalation templates. Escalated within±1 groundedness ≈0.02 vs non-escalated 1.00.

Action: rubric v2 with explicit “do not penalize abstention/escalation on groundedness”; targeted re-judge of top-15 disagreements; v1 scores preserved.

Rule: LLM judge = evaluator only — never golden/test ground truth.

Artifacts: `reports/phase3/*`, `evaluation/reply_human/judge_rubric_v{1,2}.md`, `llm_judge_scores_v1.csv`.

Status legend: IMPLEMENTED / PLANNED / OBSERVED / ASSUMED / NOT YET IMPLEMENTED